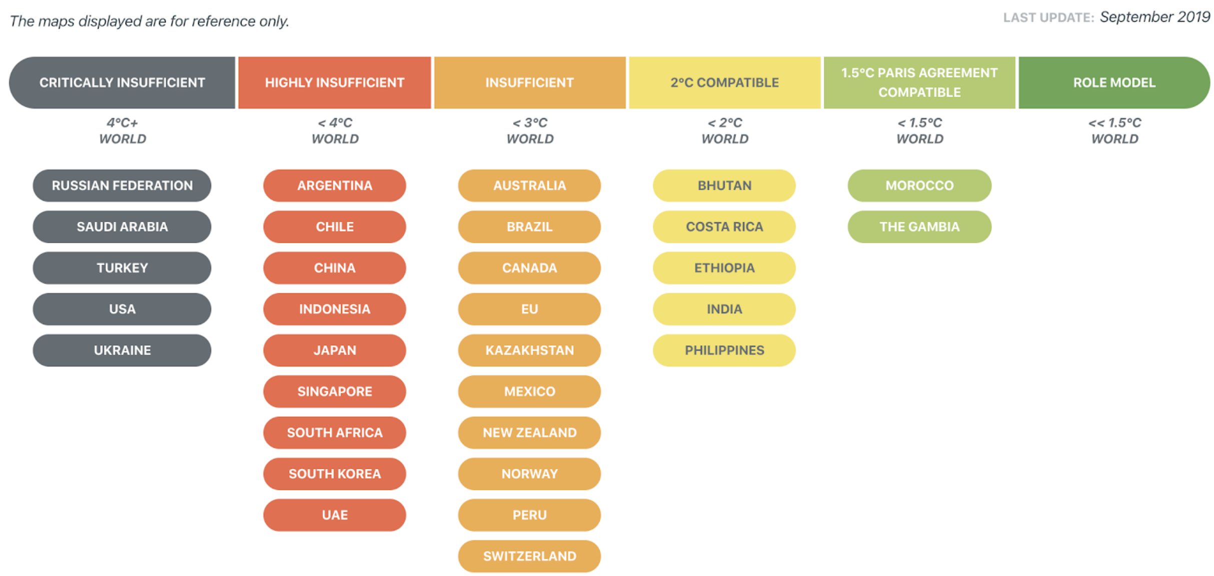
Game Theory in Explaining the Stalemate of the Climate Change Regime

The fact that scientists and governments of developed countries understood the general theory of climate change and the solution to solve it as early as the end of the 19th century might be astonishing to some of us. According to Nathaniel Rich, who writes an article about the climate change that took up a full issue of the *New York Times Magazine*, as early as 1965 the leaders of the United States knew that the climate change will lead to the rise of the sea level, the acidifying of freshwater, and even “a snowless New England, the swamping of major coastal cities, as much as a 40 percent decline in national wheat production, the forced migration of about one-quarter of the world’s population” (Rich). Nevertheless, none of the major Greenhouse Gases (GHG) emitters, countries that emit more than 2 % of the total GHG emission, including Brazil, China, India, European Union, Indonesia, Japan, Russian Federation, and United States (Luterbacher, 171), has made significant effort to reduce the global GHG emission. According to the data provided by the National Geographic and Climate Action Tracker, all of the major emitters were classified as at best “insufficient” in reducing GHG emission, the worst among them are Russian Federation and the United States; their contributions to the climate change mitigation were described as “critically insufficient” (Mulvaney). To address the climate change crisis, we did make some progress, the UN had established the international environmental treaty of climate change problem – the United Nations Framework Convention on Climate Change (UNFCCC) in 1994. Other than that, several climate agreements that require international corporation were also established alongside the UNFCCC to reduce GHG emission, but these agreements all failed because major countries failed to fulfill their commitments and international cooperation was not reached.



SOURCE: CLIMATE ACTION TRACKER

As mentioned above, since the UNFCCC took effect in 1994, it has been supplemented by multiple documents and regimes, including Tokyo Protocol in 1997, Bali action plan in 2007, followed by the 2009 Copenhagen Accord and 2010 Cancun Agreements, and finally, the Paris Agreement in 2015 (Luterbacher, 13). All of these efforts were made to ensure multilateral cooperation and participation can be reach regarding climate change and the emission of the Greenhouse Gases can be reduced. Paris Agreement was significant in terms of wide participation since there were almost 200 countries committed to the reduction of the GHG emission (Davenport). However, the Trump administration announced that the US will withdraw from the Paris Agreement because it is "job-killing" (Friedman). Although what the Trump administration did could be seen as a special case, we can still have a glimpse of underlying reasons why international cooperation is difficult to achieve and why the success of the multilateral agreements remains limited. According to Guri Bang, a senior researcher from the University of Oslo, the Paris Agreement failed because of the distrust and noncooperation between the counties. Scholars have identified the free-rider problem, which is a concept in game theory that will be explained in later paragraphs, as one of the major causes of the failure of the Paris Agreements. The same reason had led to the failure of the Kyoto Protocol. Under the Kyoto Protocol, some countries had met their commitments and successfully reduced the GHG emission, but their reduction was too small compared to the increase in the emission from other countries. The efforts made by those countries that comply with the commitments were offset by the actions taken by the other countries (Bang, 210). Political scientists like Frank Grundig, Jon Hovi, and Hugh Wardformulate suggest that the incorporation among nation-states is due to their calculation of the various factors such as economic factors, political factors, and institutional factors. For instance, a nation-state has to calculate the potential loss of economic competitiveness if they choose to reduce the emission; it might also consider political factors such as whether cutting down GHG emissions will affect the current administration's supporting rate among the electorate (Luterbacher, 94). These calculations determine whether a country would play by the rule or not; if nation-states are not given enough incentives, they will underinvest in mitigating climate change.

As mentioned above, the free-rider problem does cause the ineffectiveness of climate agreements. Scott Barrett, a professor of natural resources economics at Columbia University, has identified three requirements for a climate agreement to be effective. The first one is that agreements should have broad participation and cover major Greenhouse Gases emitters. Second, it requires countries to commit to a deep reduction in emission. Third, it requires countries to de facto implement and to achieve the emission target (Barrett, 242). Not fulfilling any one of the above requirements will result in the failure of the climate agreement, and the free-rider problem gets in the ways when nation-states deciding whether or not to fulfill these requirements. Because climate change mitigation is a global good, which means that everyone can benefits from the mitigation even without contributing to it; this characteristic of global climate change mitigation provides agents with great incentives to free-ride (Wood, 153). If the contribution to climate change mitigation is not a must to receive the benefits of it, why choose to contribute at all? The strong free-ride incentive provides explanation for the failure of the Kyoto Protocol: countries like the United States failed to ratify the commitment; countries like Canada just ignored their commitments to the reduction of GHG emission; others countries set their reduction goal extremely shallow so that they could achieve easily by contributing little (Bang, 210). Economist Mancur Olson in this book *The Logic of Collective Action: Public Goods and the Theory of Groups* suggests that there are other explanations for failures of collective action in the public goods game – that even the collective goal of the all the states are the same, the best strategy to attain that goal will depend on what other players do (Olson).

Many other economists also agree with Olson. They believe that when nation-states make their decision regarding the climate change policy, they will have to consider what other players will do and thus make the decision that is the best response to others' actions. To Michaël Aklin, a professor of political science at the University of Pittsburgh, the research interaction among states is realized by applying the game theory model (Luterbacher, 71). Not believing that other countries will contribute their part to limit the carbon emission, no policymakers will make the effort to reduce their carbon emission in the cost of their economic growth. This is also what economists called the prisoners' dilemma. According to Peter John Wood, an economist at the Australian National University, prisoners' dilemma is a term in game theory that describes a situation in which parties of the games choose to behave to maximize their interest and thus choose not to cooperate. The prisoner's dilemma usually leads to a suboptimal equilibrium which means that the maximized public social welfare is not achieved (Wood, 168). More specifically, the problem of the prisoner's dilemma is caused by the same characteristic of global climate mitigation that led to a free-rider problem – that it is a public good. Collectively, all nation-states will be better off if the global warming is slowed but each country also benefits from not contributing to the reduction of GHG emission because that means no restriction on the domestic production and industries. Wood conducts model analysis using the simplest model of a dilemma, a two players game. Each one has two choices: pollute or abate. If both players, for instance, the US and China, decide to abate, the social benefit would be maximized, and the socially optimal outcome will be achieved. If only one country chooses to abate and the other one continues to pollute, the socially optimal outcome will not be reached but the one who chooses to pollute gain more compared to the one who chooses to abate in this situation. The worst social outcome is reached when both countries choose to pollute. However, Game theory tells us that the most preferred situation for each player is not when both choose to abate but the one when it plays pollute and the other plays abates, because in this case, the one chooses to pollute enjoys the benefit brought by the one who abates without sacrificing its economic growth; the least preferred situation for each player is the one when it chooses to abate but the other one choose to pollute and free rides its efforts. In conclusion, the most preferred strategy for each player would be always choosing to pollute no matter what the other player does because given that its opponent chooses to pollute, it will be better off by choosing pollute and not sacrificing its economic growth, and if given that its opponent chooses to abate, it's in its interest to pollute and free ride. With that being said, the Nash equilibrium of the game would be both players choose to pollute, and this leads to suboptimal social welfare (Wood, 155). The "both pollute" outcome is said to be a Nash equilibrium because no one in the game would have a profitable deviation to the equilibrium, and any action that is not indicated by the equilibrium would not make the player better off. This game model shown here partially explains why superpowers like the US and China resist making the sacrifice that is necessary for the climate change mitigation, and because each country seeks to prosper, the assumption that cooperation exists under the climate agreements seems to have a shaky ground.

Although game theory is widely used as a tool to model the climate change regime, some argue that the game theory approach cannot tell the whole story. Some other aspects also play important roles in climate change policies making process but not analyzed by game theory. While these economists see interactions among nation-states as the single most important factor in the policy-making process, political scientists like James Fearon argue that domestic factors also play integrated roles in the decision making (Fearon, 577). Lobbyers and interest groups in the nation-state also involve in and can affect nation-states' behaviors on the international stage. Climate change policy is also determined by the technology factor, which in turn required domestic policy support to develop (Luterbacher, 80). The game theory approach also fails to consider how citizens of nation-states incentivize the government to act in certain ways in the global arena. As political scientists like Detlef F. Sprinz, Guri Bang, and Lars BrücknerIf would argue, the domestic politics explanation of climate change policymaking relaxes the assumption that "countries are unitary rational actors" and it concludes that "complex domestic decision-making process form the basis for a country's climate change policy and negotiation position at international negotiation tables" (Luterbacher, 176). Citizens have certain preferences for specific sets of actions that could change the position that a nation-state would take. For example, if the majority of the citizens value economic development less than the potential value of a more stable environment, then local values will also change the government's attitude to climate change and force the government to make more environmentally friendly policies.

To conclude, game theory provides two possible explanations for the failure of climate agreements like the Paris Agreement and the Kyoto Protocol, the first one is the free-rider problem and the second one is prisoner's dilemma. Both are applicable to the current stalemate in the international cooperation because of the characteristic of the climate change mitigation – that it is a public good; that any nation-states, no matter how much they reduce the GHG emission or even continue to pollute, can access the benefits of climate mitigation freely without any charge and free ride on others efforts is in their best interest. Nevertheless, when analyzing the failure of the climate change agreements, the game theory approach overlooks some of the determining factors that are involved in the process of climate change policymaking, including the domestic factors like the preferences of the citizens and local values, as well as lobbyists and interest groups. Nevertheless, whether or not it tells the whole story, game theory does give sharp prediction and plausible explanation of the current stalemate of the international efforts in climate change mitigation, and the United States withdrawal from the Paris Agreement again proved the effectiveness of the game theory approach.

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